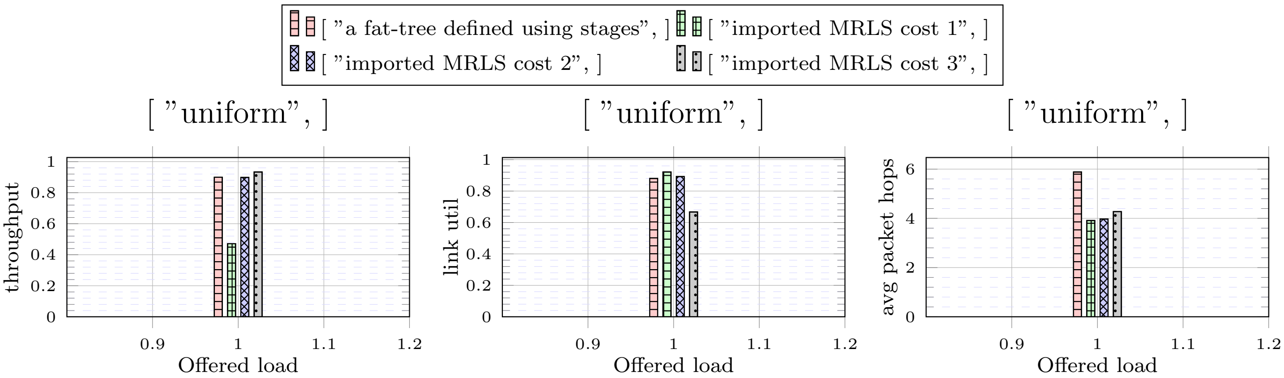


Figure 1: X: [ "uniform", ]

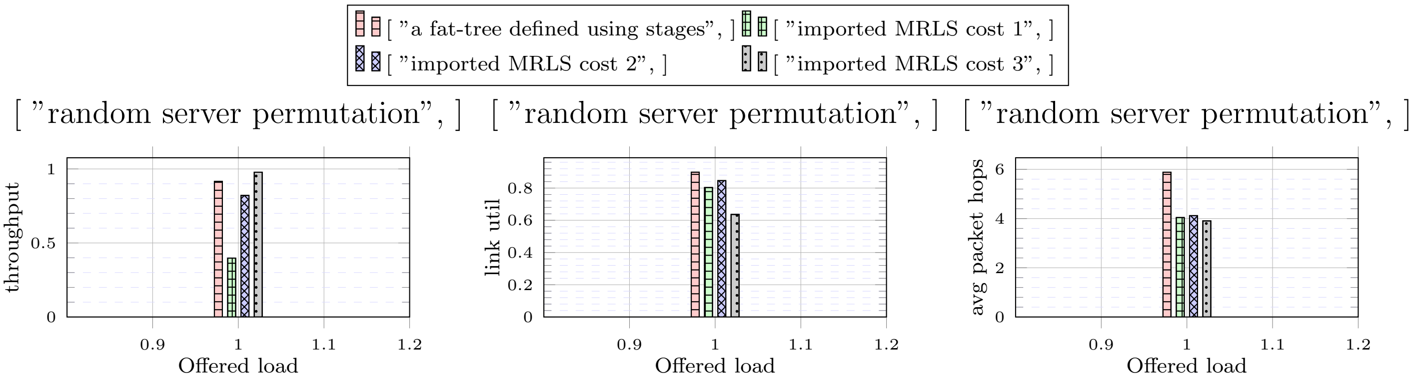


Figure 2: X: [ "random server permutation", ]

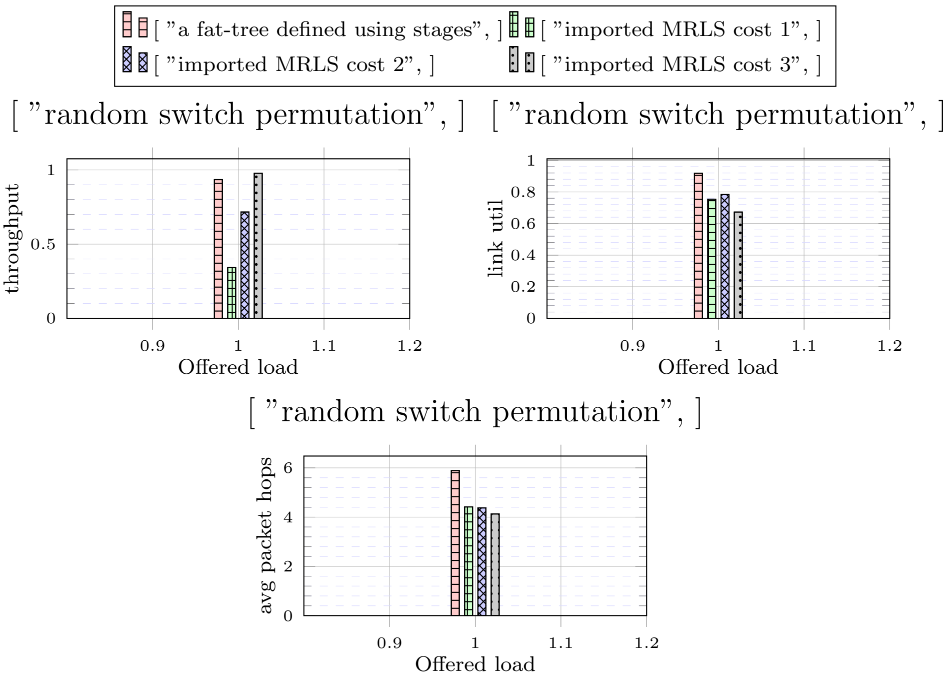


Figure 3: X: [ "random switch permutation", ]

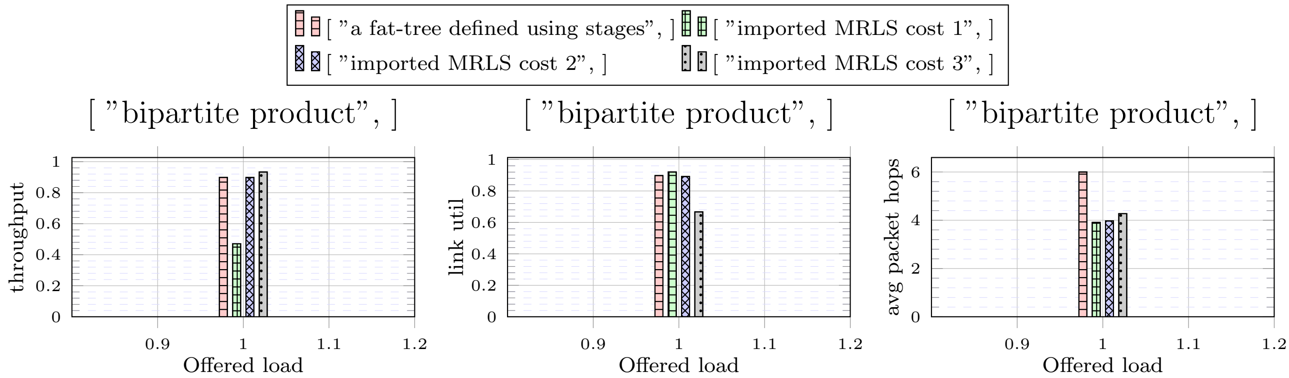


Figure 4: X: [ "bipartite product", ]

The following versions used in the simulations.

- heads/alex-stable-0ddd70b98d380a94f1ddc39d4b3992e370e55d3d(0.6.3)
- heads/alex-dev-ef489f87de796fb3ec38c1bb4368719b2cb116eb(0.6.3)
- heads/alex-dev-7775ae14e7573de7f634e7a6d77d154c320a8b9a(0.6.3)

```
Configuration{
  random_seed: ![ 42, 935, 128643 ],
  warmup: 20000,
  measured: 50000,
  general_frequency_divisor: 2,
  topology: topo![
    MultiStage{
      stages: [
        Fat{ bottom_factor: 18, top_factor: 18 },
        Fat{ bottom_factor: 18, top_factor: 18 },
        Fat{ bottom_factor: 18, top_factor: 18 }],
      servers_per_leaf: 18,
      legend_name: "a fat-tree defined using stages"},
    MultiStage{
      stages: [
        ExplicitStageFile{ filename: "../MRLS_5832_18_2916_36" }],
      servers_per_leaf: 18,
      legend_name: "imported MRLS cost 1"},
    MultiStage{
      stages: [
        ExplicitStageFile{ filename: "../MRLS_8748_24_5832_36" }],
      servers_per_leaf: 12,
      legend_name: "imported MRLS cost 2"},
    MultiStage{
      stages: [
        ExplicitStageFile{ filename: "../MRLS_11664_27_8748_36" }],
      servers_per_leaf: 9,
      legend_name: "imported MRLS cost 3"}],
  traffic: HomogeneousTraffic{
    servers: 104976,
    pattern: ![
      Uniform{ legend_name: "uniform" },
      RandomPermutation{ legend_name: "random server permutation" },
      CartesianTransform{
        sides: [ 18, 5832 ],
        patterns: [ Identity, RandomPermutation ],
        legend_name: "random switch permutation"},
      Product{
        block_size: 52488,
        block_pattern: Uniform,
        global_pattern: CartesianTransform{
          sides: [ 2 ],
          shift: [ 1 ],
          legend_name: ""},
        legend_name: "bipartite product"}],
    load: ![ 0.5 ],
    message_size: 16},
  maximum_packet_size: 16,
  router: InputOutput{
    virtual_channels: topo![ 3, 4, 4, 4 ],
    virtual_channel_policies: topo![
      [
        OccupancyFunction{ label_coefficient: 1, occupancy_coefficient: 1, product_coefficient: 0, constant_coefficient: 0, use_internal_space: true,
use_neighbour_space: true, aggregate: true },
        OccupancyFunction{ label_coefficient: 1, occupancy_coefficient: 1, product_coefficient: 0, constant_coefficient: 0, use_internal_space: true,
use_neighbour_space: true, aggregate: false },
        LowestLabel,
        EnforceFlowControl,
        Random],
      [
        MapHop{
          hop_to_policy: [
            ArgumentVC{
              allowed: [ 0 ]},
            ArgumentVC{
              allowed: [ 0 ]},
            ArgumentVC{
              allowed: [ 1 ]},
            ArgumentVC{
              allowed: [ 1 ]},
            ArgumentVC{
              allowed: [ 2 ]},
            ArgumentVC{
              allowed: [ 2 ]},
            ArgumentVC{
              allowed: [ 3 ]},
            ArgumentVC{
              allowed: [ 3 ]}]},
        VecLabel{
          label_vector: [ 0, 64, 80 ]},
        OccupancyFunction{ label_coefficient: 1, occupancy_coefficient: 1, product_coefficient: 0, constant_coefficient: 0, use_internal_space: true,
use_neighbour_space: true, aggregate: true },
        OccupancyFunction{ label_coefficient: 1, occupancy_coefficient: 1, product_coefficient: 0, constant_coefficient: 0, use_internal_space: true,
use_neighbour_space: true, aggregate: false },
        LowestLabel,
        EnforceFlowControl,
        Random],
      [
        MapHop{
          hop_to_policy: [
            ArgumentVC{
              allowed: [ 0 ]},
            ArgumentVC{
              allowed: [ 0 ]},
            ArgumentVC{
              allowed: [ 1 ]},
```

*ArgumentVC*{
**allowed**: [ 1 ]},
*ArgumentVC*{
**allowed**: [ 2 ]},
*ArgumentVC*{
**allowed**: [ 2 ]},
*ArgumentVC*{
**allowed**: [ 3 ]},
*ArgumentVC*{
**allowed**: [ 3 ]}]},
*VecLabel*{
**label_vector**: [ 0, 64, 80 ]},
*OccupancyFunction*{ **label_coefficient**: 1, **occupancy_coefficient**: 1, **product_coefficient**: 0, **constant_coefficient**: 0, **use_internal_space**: true, **use_neighbour_space**: true, **aggregate**: true },
*OccupancyFunction*{ **label_coefficient**: 1, **occupancy_coefficient**: 1, **product_coefficient**: 0, **constant_coefficient**: 0, **use_internal_space**: true, **use_neighbour_space**: true, **aggregate**: false },
*LowestLabel*,
*EnforceFlowControl*,
*Random*],
[
*MapHop*{
**hop_to_policy**: [
*ArgumentVC*{
**allowed**: [ 0 ]},
*ArgumentVC*{
**allowed**: [ 0 ]},
*ArgumentVC*{
**allowed**: [ 1 ]},
*ArgumentVC*{
**allowed**: [ 1 ]},
*ArgumentVC*{
**allowed**: [ 2 ]},
*ArgumentVC*{
**allowed**: [ 2 ]},
*ArgumentVC*{
**allowed**: [ 3 ]},
*ArgumentVC*{
**allowed**: [ 3 ]}]},
*VecLabel*{
**label_vector**: [ 0, 64, 80 ]},
*OccupancyFunction*{ **label_coefficient**: 1, **occupancy_coefficient**: 1, **product_coefficient**: 0, **constant_coefficient**: 0, **use_internal_space**: true, **use_neighbour_space**: true, **aggregate**: true },
*OccupancyFunction*{ **label_coefficient**: 1, **occupancy_coefficient**: 1, **product_coefficient**: 0, **constant_coefficient**: 0, **use_internal_space**: true, **use_neighbour_space**: true, **aggregate**: false },
*LowestLabel*,
*EnforceFlowControl*,
*Random*]],
**allocator**: *Random*,
**buffer_size**: 128,
**bubble**: false,
**flit_size**: 16,
**intransit_priority**: false,
**allow_request_busy_port**: true,
**output_buffer_size**: 64,
**crossbar_frequency_divisor**: 1,
**crossbar_delay**: 2},
**routing**: **topo!**[
*UpDown*{ **legend_name**: "minimal routing" },
*Polarized*{ **include_labels**: true, **legend_name**: "Polarized original" },
*Polarized*{ **include_labels**: true, **legend_name**: "Polarized original" },
*Polarized*{ **include_labels**: true, **legend_name**: "Polarized original" }],
**link_classes**: [
*LinkClass*{ **delay**: 2 },
*LinkClass*{ **delay**: 2 },
*LinkClass*{ **delay**: 2 },
*LinkClass*{ **delay**: 2 }],
**launch_configurations**: [
*Slurm*{ **job_pack_size**: 1, **time**: "20-00:00:00" }]}